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REFRIGERATOR AND METHOD FOR CONTROLLING REFRIGERATOR

Abstract

A refrigerator may include a main body forming a storeroom; a thermoelectric element including a heater and a cooler, the cooler configured to cool air inside the storeroom; a fan configured to operate to bring air from outside the main body to the heater to be heated by the heater and exhaust the heated air to outside the main body; a filter configured to filter foreign materials from the air brought from outside the main body to the heater by operation of the fan; and a controller configured to notify that the filter is clogged in response to a rotation speed of the fan being higher than a reference rotation speed.

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Background/Summary

[0001] This is a continuation application, under 35 U.S.C. § 111 (a), of International Application PCT/KR2025/002285, filed Feb. 17, 2025, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0032703, filed Mar. 7, 2024 and Korean Patent Application No. 10-2024-0077224, filed Jun. 13, 2024, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein in their entireties by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a refrigerator equipped with a thermoelectric element for cooling a storeroom and a method for controlling the refrigerator.

BACKGROUND ART

[0003] The refrigerator is a home appliance having a main body with storerooms and a cold air supply provided for supplying cold air into the storerooms to keep things fresh.

[0004] For the cold air supply of the refrigerator, a thermoelectric cooling device that causes heating and cooling actions through the Peltier effect may be used. The thermoelectric cooling device may include a thermoelectric element. The thermoelectric element may have a heater formed on one side and a cooler formed on the other side, and when a current is applied to the thermoelectric element, the heater may cause a heating action and the cooler may cause a heat absorption action.

[0005] The thermoelectric cooling device may be equipped with a heat sink, a cooling sink, a heat radiation fan, a cooling fan, a heat radiation duct and a cooling duct to increase cooling efficiency for the storeroom through the thermoelectric cooling device.

DISCLOSURE

Technical Problem

[0006] The disclosure provides a refrigerator and method for controlling the refrigerator, capable of notifying whether filter cleaning is completed to the user.

[0007] Technological objectives of the disclosure are not limited to what are mentioned above, and throughout the specification, it will be clearly appreciated by those of ordinary skill in the art that there may be other technological objectives unmentioned.

Technical Solution

[0008] In accordance with the present disclosure, a refrigerator may include a main body forming a storeroom; a thermoelectric element including a heater and a cooler, the cooler configured to cool air inside the storeroom; a fan configured to operate to bring air from outside the main body to the heater to be heated by the heater and exhaust the heated air to outside the main body; a filter configured to filter foreign materials from the air brought from outside the main body to the heater by operation of the fan; and a controller configured to notify that the filter is clogged in response to a rotation speed of the fan being higher than a reference rotation speed.

[0009] The controller may be further configured to: based on a lapse of a first period of time after the refrigerator is powered on, operate the fan for a second period of time, and based on the rotation speed of the fan during the second period of time, set the reference rotation speed.

[0010] The controller may be further configured to, based on an input indicating the filter is clean being received after notification that the filter is clogged, operate the fan for a third period of time.

[0011] The controller may be further configured to, based on a difference between the rotation speed of the fan during the third period of time and the reference rotation speed, determine whether the filter is clean.

[0012] The controller may be further configured to reset the reference rotation speed in response to determination that the filter is clean.

[0013] The controller may be further configured to notify that the filter is clogged in response to determination that the filter is not clean.

[0014] The reference rotation speed may include: a first reference rotation speed, and a second reference rotation speed, and the controller may be further configured to: operate the fan at a first level for the second period of time and based on the rotation speed of the fan operated at the first level, set the first reference rotation speed, operate the fan at a second level higher than the first level for the second period of time and based on the rotation speed of the fan operated at the second level, set the second reference rotation speed, notify that the filter is clogged in response to the rotation speed of the fan operated at the first level being higher than the first reference rotation speed, and notify that the filter is clogged in response to the rotation speed of the fan operated at the second level being higher than the second reference rotation speed.

[0015] The controller may be further configured to: based on an input indicating the filter is clean being received after notification that the filter is clogged, operate the fan at the first level for a third period of time and based on a difference between the rotation speed of the fan operated at the first level for the third period of time and the first reference rotation speed, determine whether the filter is clean, and operate the fan at the second level for the third period of time and based on a difference between the rotation speed of the fan operated at the second level for the third period of time and the second reference rotation speed, determine whether the filter is clean.

[0016] The thermoelectric element may be configured to operate so that a rate of temperature change of at least one of the heater and the cooler increases, and the controller may be further configured to: notify that the filter is clogged in response to: the rotation speed of the fan being higher than the reference rotation speed, and after the thermoelectric element is operated, the rate of temperature change of the at least one of the heater and the cooler is smaller than a reference value.

[0017] The refrigerator may further comprise: a communication interface configured to communicate with an external device, and the controller may be further configured to transmit information regarding the filter being clogged to the external device through the communication interface in response to the rotation speed of the fan being higher than the reference rotation speed.

[0018] In accordance with the present disclosure, a method of controlling a refrigerator including a main body forming a storeroom, a thermoelectric element including a heater and a cooler, the cooler configured to cool air inside the storeroom, a fan configured to operate to bring air from outside the main body to the heater to be heated by the heater and exhaust the heated air outside the main body, and a filter configured to filter foreign materials from the air brought from outside the main body to the heater by operation of the fan, and a controller, the method may comprise: by the controller, notifying that the filter is clogged in response to a rotation speed of the fan being higher than a reference rotation speed.

[0019] The method may further comprise: by the controller, based on a lapse of a first period of time after the refrigerator is powered on, operating the fan for a second period of time, and based on the rotation speed of the fan during the second period of time, setting the reference rotation speed.

[0020] The method may further comprise: by the controller, based on an input indicating the filter is clean being received after notification that the filter is clogged, operating the fan for a third period of time.

[0021] The method may further comprise: by the controller, based on a difference between the rotation speed of the fan during the third period of time and the reference rotation speed, determining whether the filter is clean.

[0022] The method may further comprise: by the controller, resetting the reference rotation speed in response to determining that the filter is clean.

Advantageous Effects

[0023] According to the disclosure, it is more advantageous in that whether a filter is clogged may be more accurately determined by updating reference rotation speed for determining whether the filter is clogged based on the rotation speed of a fan each time cleaning of the filter is completed.

Description

DESCRIPTION OF DRAWINGS

[0024] FIG. 1 illustrates a refrigerator, according to an embodiment of the disclosure.

[0025] FIG. 2 illustrates a refrigerator with doors open, according to an embodiment of the disclosure.

[0026] FIG. 3 illustrates an upper portion of a storeroom of a refrigerator viewed from below, according to an embodiment of the disclosure.

[0027] FIG. 4 is a schematic side cross-sectional view of a refrigerator, according to an embodiment of the disclosure.

[0028] FIG. 5 is a cross-sectional view along line I-I of FIG. 2.

[0029] FIG. 6 is a perspective view illustrating a coupling structure between a thermoelectric module and an upper wall of a refrigerator, according to an embodiment of the disclosure.

[0030] FIG. 7 is an exploded view of a heat radiation fan and a thermoelectric module, according to an embodiment of the disclosure.

[0031] FIG. 8 illustrates a first heat radiation path, a second heat radiation path and a circulation path, according to an embodiment of the disclosure.

[0032] FIG. 9 illustrates a top cover and a heat radiation duct, according to an embodiment of the disclosure.

[0033] FIG. 10 is a control block diagram of a refrigerator, according to an embodiment of the disclosure.

[0034] FIG. 11 is a conceptual diagram illustrating an example of an entity to perform a method of controlling a refrigerator, according to an embodiment of the disclosure.

[0035] FIG. 12 illustrates an example of a flowchart of a method of controlling a refrigerator, according to an embodiment of the disclosure.

[0036] FIG. 13 illustrates an example of an interface provided by a refrigerator or a user device, according to an embodiment of the disclosure.

[0037] FIG. 14 illustrates another example of an interface provided by a refrigerator or a user device, according to an embodiment of the disclosure.

[0038] FIG. 15 illustrates an example of an interface provided by the refrigerator or the user device when a user input is received through the interface of FIG. 13 or 14.

[0039] FIG. 16 illustrates an example of a flowchart of a method of controlling a refrigerator, according to an embodiment of the disclosure.

[0040] FIG. 17 illustrates an example of a flowchart of a method of controlling a refrigerator, according to an embodiment of the disclosure.

[0041] FIG. 18 illustrates an example of a flowchart of a method of controlling a refrigerator for notifying filter clogging based on rotation speed of a fan and temperature of a thermoelectric element, according to an embodiment of the disclosure.

MODES OF THE INVENTION

[0042] It is understood that various embodiments of the disclosure and associated terms are not intended to limit technical features herein to particular embodiments, but encompass various changes, equivalents, or substitutions.

[0043] Like reference numerals may be used for like or related elements throughout the drawings.

[0044] The singular form of a noun corresponding to an item may include one or more items unless the context states otherwise.

[0045] Throughout the specification, “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B or C”, “at least one of A, B and C”, and “at least one of A, B, or C” may each include any one or all the possible combinations of A, B and C.

[0046] The expression “and/or” is interpreted to include a combination or any of associated elements.

[0047] A unit, module or member may be implemented in hardware or software. A plurality of units, modules or members may be implemented as one component, or a single unit, module or member may include multiple components.

[0048] Terms like “first”, “second”, etc., may be simply used to distinguish an element from another, without limiting the elements in a certain sense (e.g., in terms of importance or order).

[0049] When an element is mentioned as being “coupled” or “connected” to another element with or without an adverb “functionally” or “operatively”, it means that the element may be connected to the other element directly (e.g., wiredly), wirelessly, or through a third element.

[0050] It will be further understood that the terms “comprise” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, parts or combinations thereof, but do not preclude the possible presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0051] When an element is mentioned as being “connected to”, “coupled to”, “supported on” or “contacting” another element, it includes not only a case that the elements are directly connected to, coupled to, supported on or contact each other but also a case that the elements are connected to, coupled to, supported on or contact each other through a third element.

[0052] Throughout the specification, when an element is mentioned as being located “on” another element, it implies not only that the element is abut on the other element but also that a third element exists between the two elements.

[0053] The terms “forward or front”, “rearward or back”, “left”, “right”, “upper or up” or “lower or down” as herein used are defined with respect to the drawings, but the terms may not restrict the shapes and position of the respective components. For example, the front may be defined as +X direction and the back may be defined as -X direction. For example, with respect to the drawings, the right may be defined as +Y direction and the left may be defined as -Y direction. For example, with respect to the drawings, the upper direction may be defined as +Z direction and the lower direction may be defined as -Z direction.

[0054] Reference will now be made in detail to embodiments of the disclosure, which are illustrated in the accompanying drawings.

[0055] In an embodiment, a refrigerator may include a main body.

[0056] The main body may include insulation. The insulation may insulate inside and outside of the storeroom so that the temperature in the storeroom is maintained at a set suitable temperature without being influenced by external environments of the storeroom. In an embodiment, the insulation may include foam insulation such as polyurethane foam. In an embodiment, the insulation may include an additional vacuum insulation in addition to the foam insulation, or may include only the vacuum insulation instead of the foam insulation.

[0057] Various items such as foods, medicines, cosmetics, etc., may be stored in the storeroom, and the storeroom may be formed with one side open to put in or take out the items.

[0058] The refrigerator may include one or more storerooms. When there are two or more storerooms formed in the refrigerator, each storeroom may have a different use and may be

maintained at a different temperature. For this, the storerooms may be separated by partition walls including insulation.

[0059] The storerooms may be provided to be each maintained at a suitable temperature range, and may include a fridge, a freezer or a temperature-changing room classified by the use and/or the temperature range. The fridge may be maintained at a suitable temperature for keeping items refrigerated, and the freezer may be maintained at a suitable temperature for keeping items frozen. Refrigeration may refer to cooling the items to an extent that the items are not frozen, and for example, the fridge may be maintained at a range of 0 to 7 degrees Celsius above zero. Freezing may refer to freezing the items or cooling the items in a frozen state, and for example, the freezer may be maintained at a range of 20 to 1 degree Celsius below zero. The temperature-changing room may be used as one of the fridge or the freezer according to or regardless of the user's choice.

[0060] The storerooms may be called many different names such as “veggie room”, “fresh room”, “cooling room” and “ice-making room” in addition to the names such as “fridge chamber”, “freezer chamber” and “temperature-changing room”, and the terms such as “fridge chamber”, “freezer chamber” and “temperature-changing room” need to be understood as encompassing storerooms having respective uses and temperature ranges.

[0061] In an embodiment, the refrigerator may include at least one door arranged to open or close the one open side of the storeroom. A door may be equipped to open or close each of the one or more storerooms, or one door may be equipped to open or close multiple storerooms. The door may be rotationally or slidably installed at the front of the main body.

[0062] The door may be arranged to close the storeroom tight when closed. Like the main body, the door may include insulation to insulate the storeroom when closed.

[0063] In an embodiment, the door may include a door outer-plate that forms the front surface of the door, a door inner-plate that forms the rear surface of the door and faces the storeroom, an upper cap, a lower cap and door insulation provided inside of them.

[0064] A gasket may be arranged on edges of the door inner-plate to seal the storeroom by tightly contacting the front surface of the main body when the door is closed. The door inner-plate may include a dyke that protrudes rearward for a door basket that may keep items to be installed thereon.

[0065] In an embodiment, the door may include a door body and a front panel detachably coupled to the front side of the door body and forming the front of the door. The door body may include a door outer-plate that forms the front surface of the door body, a door inner-plate that forms the rear surface of the door body and faces the storeroom, an upper cap, a lower cap and door insulation provided inside of them.

[0066] The refrigerator may be distinguished according to the layout of the door(s) and storeroom(s) as a French door type, a side-by-side type, a bottom mounted freezer (BMF), a top mounted freezer (TMF) or a one-door refrigerator.

[0067] In an embodiment, the refrigerator may include a cold air supplier arranged to supply cold air into the storeroom.

[0068] The cold air supplier may include a machine, instrument, electronic device and/or system that combines them, which is able to produce and lead cold air to cool the storeroom.

[0069] In an embodiment, the cold air supplier may produce the cold air through a refrigeration cycle including processes of compression, condensation, expansion and evaporation of a refrigerant. For this, the cold air supplier may include a refrigeration cycle system having a compressor, a condenser, an expansion device and an evaporator that may operate the refrigeration cycle. In an embodiment, the cold air supplier may include a semiconductor such as a thermoelectric element. The thermoelectric element may cool the storeroom by heating and cooling actions through the Peltier effect.

[0070] In an embodiment, the refrigerator may include a machine room arranged for at least some parts belonging to the cold air supplier to be placed therein.

[0071] The machine room may be separated and insulated from the storeroom to prevent heat generated from the parts arranged in the machine room from being transferred to the storeroom. To emit heat from the parts arranged in the machine room, the inside of the machine room may be formed to connect to the outside of the main body.

[0072] In an embodiment, the refrigerator may include a dispenser arranged at the door to provide water and/or ice. The dispenser may be located at the door for the user to make access thereto without a need to open the door.

[0073] In an embodiment, the refrigerator may include an ice maker provided to produce ice. The ice maker may include an ice maker tray for storing water, an ice separator for separating ice from the ice maker tray, and an ice bucket for storing ice produced from the ice maker tray.

[0074] In an embodiment, the refrigerator may include a controller for controlling the refrigerator.

[0075] The controller may include a memory for storing or memorizing a program and/or data for controlling the refrigerator, and a processor for outputting control signals to control components such as the cold air supplier according to the program and/or data stored in the memory.

[0076] The memory stores or records various information, data, instructions, programs, etc., required for operation of the refrigerator. The memory may store temporary data that is generated while the control signals to control the components included in the refrigerator are being generated. The memory may include at least one or a combination of volatile memories or non-volatile memories.

[0077] The processor controls general operation of the refrigerator. The processor may control the components of the refrigerator by executing the program stored in the memory. The processor may include an extra neural processing unit (NPU) that performs operation of an artificial intelligence (AI) model. The processor may also include a central processing unit (CPU), a graphic processing unit (GPU), etc. The processor may generate control signals to control operation of the cold air supplier. For example, the processor may receive information about the temperature in the storeroom from a temperature sensor, and generate a refrigeration control signal to control an operation of the cold air supplier based on the temperature information of the storeroom.

[0078] Furthermore, the processor may process a user input to a user interface according to the program and/or data memorized/stored in the memory, and control operation of the user interface. The user interface may be provided by using an input interface and an output interface. The processor may receive the user input from the user interface. Furthermore, the processor may send a display control signal and image data for displaying an image on the user interface to the user interface in response to the user input.

[0079] The processor and the memory may be provided in one unit or separately. The processor may include one or more processors. For example, the processor may include a main processor and at least one subprocessor. The memory may include one or more memories.

[0080] In an embodiment, the refrigerator may include a processor and a memory for controlling all the components included in the refrigerator, or include a plurality of processors and a plurality of memories for controlling the components of the refrigerator, respectively. For example, the refrigerator may include a processor and a memory for controlling operation of the cold air supplier according to the output of the temperature sensor. The refrigerator may include another processor and another memory for controlling operation of the user interface according to the user input.

[0081] A communication module may communicate with an external device such as a server, a mobile device, another home appliance, etc., through a nearby access point (AP). The AP may connect a local area network (LAN) connected to the refrigerator or user device to a wide area network (WAN) connected to the server. The refrigerator or user device may be connected to the server through the WAN.

[0082] The input interface may include a key, a touch screen, a microphone, etc. The input interface may receive a user input and forward it to the processor.

[0083] The output interface may include a display, a speaker, etc. The output interface may output

various notifications, alert, messages, information, etc., generated by the processor.

[0084] In the disclosure, operating an electronic part may include turning on the electronic part. In the disclosure, operating an electronic part may include maintaining an on state of the electronic part.

[0085] A working principle and embodiments of the disclosure will now be described with reference to accompanying drawings.

[0086] FIG. 1 illustrates a refrigerator, according to an embodiment of the disclosure. FIG. 2 illustrates a refrigerator with doors open, according to an embodiment of the disclosure. FIG. 3 illustrates an upper portion of a storeroom of a refrigerator viewed from below, according to an embodiment of the disclosure. FIG. 4 is a schematic side cross-sectional view of a refrigerator, according to an embodiment of the disclosure. FIG. 5 is a cross-sectional view along line I-I of FIG. 2.

[0087] Referring to FIG. 1, a refrigerator 1 may include a main body 100, storerooms 11, 12, and 13 formed in the main body 100, and doors 21, 22, 23, and 24 arranged to open or close the storerooms 11, 12, and 13. The doors 21, 22, 23 and 24 may include the first door 21, the second door 22, the third door 23 and/or the fourth door 24.

[0088] The refrigerator 1 may include a user interface device 260. The user interface device 260 may be installed on at least one of the doors 21, 22, 23 and 24. For example, the user interface device 260 may be installed on the second door 22 among the doors 21, 22, 23 and 24. However, the location on the refrigerator 1 at which the user interface device 260 is installed is not limited thereto, and the user interface device 260 may be installed at various locations on the refrigerator 1. For example, the user interface device 260 may be installed on a right wall 140.

[0089] Referring to FIGS. 1 to 5, the main body 100 may include an upper wall 110, a lower wall 120, a left wall 130, a right wall 140 and a rear wall 150. The upper wall 110, the lower wall 120, the left wall 130, the right wall 140 and the rear wall 150 may define the upper surface, lower surface, left surface, right surface and rear surface of the main body 100, respectively.

[0090] The storerooms 11, 12 and 13 may accommodate items. The storerooms 11, 12 and 13 may be formed to have open front to put in or take out the items. The main body 100 may include a horizontal partition wall 160 that divides a first storeroom 11 from a second storeroom 12 and a third storeroom 13, and a vertical partition wall 161 that divides the second storeroom 12 from the third storeroom 13. The first storeroom 11 may be arranged in an upper portion of the main body 100, and the second storeroom 12 and the third storeroom 13 may be arranged in a lower portion of the main body 100. The first storeroom 11 may be a fridge chamber; the second storeroom 12 may be a freezer chamber; the third storeroom 13 may be a temperature-changing chamber.

[0091] The doors 21, 22, 23 and 24 may open or close the storerooms 11, 12 and 13. The first door 21 and the second door 22 may open or close the first storeroom 11, the third door 23 may open or close the second storeroom 12, and the fourth door 24 may open or close the third storeroom 13. The doors 21, 22, 22 and 24 may be rotationally coupled to the main body 100.

[0092] The doors 21, 22, 22 and 24 may be rotationally coupled to the main body 100 by hinges. For example, the first door 21 and the second door 22 may be rotationally coupled to the main body 100 by hinges 31 arranged at the top of the main body 100 and hinges arranged in the middle of the main body 100. The hinge 31 may include a hinge pin that vertically protrudes to form a rotation shaft of the door. The hinge 31 may be covered by a top cover 300 arranged to cover the top front portion of the main body 100.

[0093] A rotation bar 40 may be arranged at one of the first door 21 and the second door 22 to cover a gap formed between the first door 21 and the second door 22 while the first door 21 and the second door 22 are closed. The rotation bar 40 may be rotationally arranged at one of the first door 21 and the second door 22. The rotation bar 40 may have the shape of a bar formed to be long in the vertical direction. The rotation bar 40 may also be referred to as a pillar or a mullion.

[0094] A guide projection 46 may be arranged at the top of the rotation bar 40, and a rotation guide

119 for guiding rotation of the guide projection **46** may be arranged at the top of the main body **100**.

[0095] The doors **21**, **22**, **23** and **24** may include gaskets **51**. The gaskets **51** may closely come into contact with the front surface of the main body **100** while the doors **21**, **22**, **23** and **24** are closed. The doors **21**, **22**, **23** and **24** may include dikes **52** that protrude rearward. Door racks **53** may be mounted on the dike **52** to store items. The rotation bar **40** may be rotationally installed at the dike **52**.

[0096] Although the number and layout of storerooms and the number and layout of doors were described above, there are no limitations on the number and layout of storerooms and the number and layout of doors of the refrigerator according to an embodiment of the disclosure.

[0097] The refrigerator **1** may include a thermoelectric cooling device **400** arranged to cool the storeroom **11**.

[0098] The thermoelectric cooling device **400** may be arranged above the storeroom **11** to cool the storeroom **11**. Specifically, the thermoelectric cooling device **400** may be arranged at the upper wall **110** of the main body **100**.

[0099] The thermoelectric cooling device **400** may include a thermoelectric element **530**. The thermoelectric element **530** may be a semiconductor device that uses the thermoelectric effect to convert thermal energy to electric energy and vice versa, and may also be referred to as a thermoelectric device, a Peltier element, etc.

[0100] The thermoelectric element **530** includes a heater **531** and a cooler **532**. When a current is applied to the thermoelectric element **530**, a heat radiation action may occur in the heater **531** and a heat absorption action may occur in the cooler **532**. The thermoelectric element **530** may be shaped like a thin hexahedron. The heater **531** may be arranged on one side of the thermoelectric element **530** and the cooler **532** may be arranged on the other side.

[0101] The thermoelectric element **530** may be arranged such that the heater **531** is directed upward of the thermoelectric element **530** and the cooler **532** is directed downward of the thermoelectric element **530**. In other words, the heater **531** may face outside of the main body **100** and the cooler **532** may face the inside of the storeroom **11**. Accordingly, the air heated by exchanging heat with the heater **531** may be discharged out of the main body **100** and the air cooled by exchanging heat with the cooler **532** may be supplied into the storeroom **11**.

[0102] The thermoelectric cooling device **400** may include a heat radiation sink **520** that comes into contact with the heater **531** to efficiently perform heat exchange between the heater **531** and the air outside the main body **100**.

[0103] The heat radiation sink **520** may be located outside the main body **100**. The heat radiation sink **520** may come into contact with the heater **531** to absorb heat from the heater **531** and emit heat to the outside of the main body **100**. The heat radiation sink **520** may also be referred to as a hot sink, a heat sink for heat dissipation, a hot heat sink, etc.

[0104] The heat radiation sink **520** may be formed of a metal with high heat conductivity. For example, the heat radiation sink **520** may be formed of aluminum or copper.

[0105] The heat radiation sink **520** may include a heat radiation sink base **521** that comes into contact with the heater **531**, and a plurality of heat radiation pins **525** protruding from the heat radiation sink base **521** to expand the heating surface. The plurality of heat sink pins **525** may protrude upward from the heat radiation sink base **521**.

[0106] The thermoelectric cooling device **400** may include a cooling sink **570** that comes into contact with the cooler **532** to efficiently perform heat exchange between the cooler **532** and the air inside the storeroom **11**.

[0107] The cooling sink **570** may be located in the storeroom **11**. The cooling sink **570** may cool the storeroom **11** by taking heat from the storeroom **11** and transferring the heat to the cooler **532**. The cooling sink **570** may also be referred to as a cold sink, a refrigeration sink, a cold heat sink, a cooling heat sink, etc.

[0108] The cooling sink **570** may be formed of a metal with high heat conductivity. For example, the cooling sink **570** may be formed of aluminum or copper.

[0109] The cooling sink **570** may include a cooling sink base **571** that comes into contact with the cooler **532**, and a plurality of cooling pins **575** protruding from the cooling sink base **571** to expand the heating surface. The plurality of cooling pins **525** may protrude downward from the cooling sink base **571**. The cooling sink base **571** and the plurality of cooling pins **575** may be integrally formed.

[0110] The thermoelectric cooling device **400** may include a heat radiation fan **600** that forces air to move to efficiently perform heat exchange between the heat radiation sink **520** and the air outside the main body **100**.

[0111] The heat radiation fan **600** may be arranged to blow air to the heat radiation sink **520**. The heat radiation fan **600** may be located in a horizontal direction of the heat radiation sink **520**. The heat radiation fan **600** may be arranged outside the main body **100**. The heat radiation fan **600** may be arranged on the top of the upper wall **110**.

[0112] The heat radiation fan **600** may be a centrifugal fan that draws air in the axial direction and discharges the air in the radial direction. The centrifugal fan may include a blower fan. A rotation shaft **610** of the heat radiation fan **600** may be arranged to be perpendicular to the top surface of the upper wall **110**.

[0113] The thermoelectric cooling device **400** may include a heat radiation duct **700** arranged to guide the air moving by the heat radiation fan **600**. The heat radiation duct **700** may draw in air from outside the main body **100** and guide the air to exchange heat with the heat radiation sink **520**, and discharge the air that has exchanged heat with the heat radiation sink **520** back to the outside of the main body **100**.

[0114] The heat radiation duct **700** may draw in air from an outer space above the main body **100**. The heat radiation duct **700** may discharge the air that has exchanged heat with the heat radiation sink **520** to the outer space above the main body **100**. The heat radiation fan **600** may be located in the heat radiation duct **700**. The heat radiation sink **520** may be located in the heat radiation duct **700**. The heat radiation duct **700** may be arranged on the top surface of the upper wall **110**.

[0115] The heat radiation duct **700** may include an outside air intake port **751** through which to draw in air from outside the main body **100** into the heat radiation duct **700**, and an outside air discharge port **782** through which to discharge the air that has exchanged heat with the heat radiation sink **520** to the outside of the main body **100**.

[0116] The thermoelectric cooling device **400** may include a cooling fan **800** that forces air to move to efficiently perform heat exchange between the cooling sink **570** and the air inside the storeroom **11**.

[0117] The cooling fan **800** may be arranged to blow air to the cooling sink **570**. The cooling fan **800** may be located in a horizontal direction of the cooling sink **570**. The cooling fan **800** may be arranged in the storeroom **11**. The cooling fan **800** may be arranged underneath the upper wall **110**.

[0118] The cooling fan **800** may be a centrifugal fan that draws air in the axial direction and discharges the air in the radial direction. A rotation shaft **810** of the cooling fan **800** may be arranged to be perpendicular to the bottom surface of the upper wall **110**.

[0119] The thermoelectric cooling device **400** may include a cooling duct **900** arranged to guide the air moving by the cooling fan **800**. The cooling duct **700** may draw in air from inside the storeroom **11** and guide the air to exchange heat with the cooling sink **570**, and discharge the air that has exchanged heat with the cooling sink **570** back into the storeroom **11**.

[0120] The cooling fan **800** may be located in the cooling duct **900**. The cooling sink **570** may be located in the cooling duct **900**. The cooling duct **900** may be arranged underneath the upper wall **110**.

[0121] The cooling duct **900** may include an inside air intake port **991** through which to draw in air from inside the storeroom **11** into the cooling duct **900**, and an inside air discharge port **992** through

which to discharge the air that has exchanged heat with the cooling sink **570** into the storeroom **11**.
[0122] Referring to FIG. **4**, the refrigerator **1** may include a refrigeration cycle device to cool the storeroom through a refrigeration cycle. The refrigeration cycle device may include a compressor **2**, a condenser (not shown), an expansion valve (not shown) and an evaporator **3**. The evaporator **3** may be arranged beside the storerooms **12** and **13**.

[0123] The refrigerator **1** may include evaporator ducts **60** and **70** that guide cold air produced from the evaporator **3**. The first evaporator duct **60** may be arranged behind the second storeroom **12** and the third storeroom **13**. The second evaporator duct **70** may be arranged behind the first storeroom **11**.

[0124] The cold air produced from the evaporator **3** may be drawn into the first evaporator duct **60** by the evaporator fan **80**. The cold air drawn into the first evaporator duct **60** may be discharged into the second storeroom **12** or the third storeroom **13** through a cold air outlet (not shown) formed on the front. Furthermore, the cold air drawn into the first evaporator duct **60** may be guided into an internal flow path **78** of the second evaporator duct **70**. A damper **61** may be arranged in the first evaporator duct **60** to control the cold air in the first evaporator duct **60** to be supplied into the second evaporator duct **70**. A connection duct **90** may be arranged between the first evaporator duct **60** and the second evaporator duct **70** to connect the first evaporator duct **60** to the second evaporator duct **70**.

[0125] The cold air brought into the internal flow path **78** of the second evaporator duct **70** may be supplied into the first storeroom **11** through a cold air outlet **72** formed on the front of the second evaporator duct **70**.

[0126] The damper **61** may open or close the internal flow path **78**.

[0127] When the internal flow path **78** is opened by the damper **61**, the cold air produced from the evaporator **3** may be guided into the first storeroom **11**.

[0128] When the internal flow path **78** is closed by the damper **61**, the cold air produced from the evaporator **3** may be blocked by the damper **61** and may not be guided into the first storeroom **11**.

[0129] The cold air brought into the internal flow path **78** of the second evaporator duct **70** may be supplied into the first storeroom **11** through the cold air outlet **72** formed on the front of the second evaporator duct **70**.

[0130] However, unlike the aforementioned embodiment, the cold air produced from the evaporator **3** may be supplied directly into the second evaporator duct **70** without passing through the first evaporator duct **60**. Furthermore, the separate evaporator **3** may be arranged behind the first storeroom **11** to supply cold air into the second evaporator duct **70**.

[0131] As such, as the refrigerator **1** according to an embodiment of the disclosure includes the thermoelectric cooling device **400** and the refrigeration cycle device for cooling the storeroom **11**, the method of supplying cold air into the storeroom **11** may include a first method of supplying cold air produced only by the thermoelectric cooling device **400**, a second method of supplying cold air produced only by the refrigeration cycle device, and a third method of supplying cold air produced by both the thermoelectric cooling device **400** and the refrigeration cycle device.

[0132] The refrigerator **1** may supply cold air into the storeroom **11** in a proper method according to external and internal conditions. For example, the refrigerator **1** may cool the storeroom **1** in one of the methods according to the temperature of a room where the refrigerator **1** is installed. Specifically, when the room temperature is higher than a predetermined temperature, so cooling by the refrigeration cycle has higher efficiency than by thermoelectric cooling device **400**, the storeroom **11** may be cooled by the cold air produced only by the refrigeration cycle device. On the other hand, when the room temperature is lower than the predetermined temperature, so cooling by the thermoelectric cooling device **400** has higher efficiency than by the refrigeration cycle device, the storeroom **11** may be cooled by the cold air produced only by the thermoelectric cooling device **400**. The refrigerator **1** may operate only the thermoelectric cooling device **400** when there is a need to reduce noise. The refrigerator **1** may supply cold air produced by the thermoelectric cooling

device **400** and cold air produced by the refrigeration cycle device into the storeroom **11** at the same time when the storeroom **11** needs to be cooled rapidly.

[0133] As described above, the refrigerator **1** according to an embodiment of the disclosure may include the thermoelectric cooling device **400** and the refrigeration cycle device, but is not limited thereto and the refrigerator may include only the thermoelectric cooling device **400**.

[0134] Although the thermoelectric cooling device **400** was described as being arranged at the upper wall **110** of the main body **100**, the location of the thermoelectric cooling device **400** is not limited thereto.

[0135] In various embodiments, the thermoelectric cooling device **400** may be arranged at at least one of the upper wall **110**, the lower wall **120**, the left wall **130**, the right wall **140** and the rear wall **150**.

[0136] The refrigerator **1** may include a first speed sensor **111** and/or a second speed sensor **112**.

[0137] The first speed sensor **111** may be arranged in the heat radiation duct **700**. The first speed sensor **111** may detect rotation speed of the heat radiation fan **600**.

[0138] The second speed sensor **112** may be arranged in the cooling duct **900**. The second speed sensor **112** may detect rotation speed of the cooling fan **800**.

[0139] The refrigerator **1** may include a first temperature sensor **113** and/or a second temperature sensor **114**.

[0140] The first temperature sensor **113** may detect the temperature of the heater **531**.

[0141] The second temperature sensor **114** may detect the temperature of the cooler **532**.

[0142] FIG. **6** is a perspective view illustrating a coupling structure between a thermoelectric module and an upper wall of a refrigerator, according to an embodiment of the disclosure. FIG. **7** is an exploded view of a heat radiation fan and a thermoelectric module, according to an embodiment of the disclosure.

[0143] The thermoelectric element **530**, the heat radiation sink **520** and the cooling sink **570** among the components of the thermoelectric cooling device may be assembled integrally into a thermoelectric module.

[0144] A through hole **115** may be formed at the upper wall **110** of the refrigerator **1**, and the thermoelectric module may be arranged in the through hole **115**.

[0145] The thermoelectric module may include the thermoelectric element **530** having the heater **531** and the cooler **532**, the heat radiation sink **520** that comes into contact with the heater **531** of the thermoelectric element **530**, the cooling sink **570** that comes into contact with the cooler **532** of the thermoelectric element **530**, and a module plate **550** on which the thermoelectric element **530**, the heat radiation sink **520** and the cooling sink **570** are installed.

[0146] The module plate **550** may serve as a frame of the thermoelectric module. The module plate **550** may be formed of a resin material with low heat conductivity. The module plate **550** may support the heat radiation sink **520** and the cooling sink **570**. The module plate **550** may keep a distance between the heat radiation sink **520** and the cooling sink **570**, and support the heat radiation sink **520** and the cooling sink **570**. As shown in the drawings, the module plate **550** may be integrally formed with a fan case **650** which will be described later. However, it is also possible that the module plate **550** is arranged separately from the fan case **650**.

[0147] The module plate **550** may include a module plate opening **551**. The thermoelectric element **530** may be arranged inside the module plate opening **551**. The vertical length of the module plate opening **551** may be larger than the vertical length of the thermoelectric element **530**, and the thermoelectric element **530** may be arranged on the top of the module plate opening **551**. The reason why the thermoelectric element **530** is arranged on the top of the inside of the module plate opening **551** is that an amount of heat radiation of the thermoelectric element **530** is larger than an amount of heat absorption, that the thermoelectric element **530** being located on the top of the module plate opening **551** is advantageous to heat radiation of the heater **531** and that overall operation efficiency of the thermoelectric element **530** may increase.

[0148] As the thermoelectric element **530** is arranged on the top of the module plate opening **551**, the cooling sink **570** may include a cooling conductor **574** that protrudes from the cooling sink base **571** to come into contact with the cooler **532** of the thermoelectric element **530**. The cooling conductor **574** may be inserted to the module plate opening **551** to come into contact with the cooler **532** of the thermoelectric element **530**.

[0149] The thermoelectric module may include an element insulator **540** that insulates the module plate **550** from the thermoelectric element **530**. The element insulator **540** may be arranged on the module plate opening **551** to prevent the thermoelectric element **530** from contacting the module plate **550**. The element insulator **540** may be arranged to surround the sides of the thermoelectric element **530**. The element insulator **540** may include an element insulator opening **541** so that the thermoelectric element **530** may be accommodated in the element insulator opening **541**. The cooling conductor **574** of the cooling sink **570** may be accommodated in the element insulator opening **541**.

[0150] The module plate **550** may include a heat radiation sink support **552** for supporting the heat radiation sink **520**. The heat radiation sink support **552** may support by coming into contact with the bottom surface of the heat radiation sink base **521**.

[0151] The heat radiation module may include a sink insulator **580** arranged between the module plate **550** and the cooling sink **570**. The sink insulator **580** may prevent heat transfer between the heat radiation sink **520** and the cooling sink **570** through the module plate **550**. The sink insulator **580** may include a sink insulator opening **581** for the opening **551** of the module plate **550** to pass through.

[0152] The sink insulator **580** may support the top of the cooling sink **570**. However, the sink insulator **580** may be omitted, and in this case, the heat radiation sink **520** may be supported on the top of the module plate **550** and the cooling sink **570** may be supported on the bottom of the module plate **550**.

[0153] The heat radiation sink **520** may include the heat radiation sink base **521** and the plurality of heat radiation pins **525** protruding from the heat radiation sink base **521**. The bottom of the heat radiation sink base **521** may be supported on the module plate **550**. The bottom of the heat radiation sink base **521** may contact the heater **531** of the thermoelectric element **530**.

[0154] The plurality of heat sink pins **525** may protrude from the top of the heat radiation sink base **521**.

[0155] The cooling sink **570** may include the cooling sink base **571**, and the plurality of cooling pins **575** protruding from the cooling sink base **571**. The top of the cooling sink base **571** may be supported by the sink insulator **580**. The cooling sink **570** may include the cooling conductor **574** that protrudes from the top of the cooling sink base **571** to come into contact with the cooler **532** of the thermoelectric element **530**. The cooling conductor **574** may be formed integrally with the cooling sink base **571**.

[0156] The plurality of cooling pins **575** may protrude from the bottom **572** of the cooling sink base **571**.

[0157] The heater **531** of the thermoelectric element **530** may be fixed by being supported by the heat radiation sink **520**; the cooler **532** of the thermoelectric element **530** may be fixed by being supported by the cooling sink **570**; the side that connects the heater **531** and the cooler **532** of the thermoelectric element **530** may be fixed by being supported by inner sides of the element insulator **540**.

[0158] The refrigerator **1** may include the fan case **650** in which the heat radiation fan **600** is installed to guide air blown by the heat radiation fan **600**. The fan case **650** may be formed integrally with or separately from the aforementioned module plate **550**.

[0159] The fan case **650** may include a case bottom **660** on which the heat radiation fan **600** is rotationally installed, and a case scroll **670** extending upward from edges of the case bottom **660** to guide air blown from the heat radiation fan **600** toward the heat radiation sink **520**. The heat

radiation fan **600** may be a centrifugal fan, which may be installed to be perpendicular to the case bottom **660**.

[0160] The case scroll **670** may be formed to surround the heat radiation fan **600** in the radial direction of the heat radiation fan **600** and have a scroll opening **673** open to the heat radiation sink **520**. The case scroll **670** may include one end **671** in a rotation direction R of the heat radiation fan **600** and the other end **672** opposite the one end. The one end **671** and the other end **672** may be separated from each other, and the scroll opening **673** may be formed between the one end **671** and the other end **672**.

[0161] The air blown from the heat radiation fan **600** may be discharged in the radial direction of the heat radiation fan **600** and moved to the heat radiation sink **520** along inner sides of the case scroll **670**. More air may be moved from the heat radiation fan **600** toward the one end **671** of the case scroll **670** than toward the other end **672** of the case scroll **670**.

[0162] The fan case **650** may include a case guide **680** extending upward from the case bottom **660** to guide the air moving toward the one end **671** of the case scroll **670** from the heat radiation fan **600** to a different direction. The case guide **680** may be spaced from the case scroll **670**. The case guide **680** may guide the air moving toward the one end **671** of the case scroll **670** to the other end **672** of the case scroll **670** or to the center of the case opening **673**. Accordingly, the air blown from the heat radiation fan **600** may be uniformly distributed by the case guide **680** to heat radiation channels **528** of the heat radiation sink **520**, and the heat exchange efficiency of the heat radiation sink **620** may increase.

[0163] Symbol S in FIG. **6** denotes coupling members S to couple the thermoelectric module to the upper wall **110**.

[0164] FIG. **8** illustrates a first heat radiation path, a second heat radiation path and a circulation path, according to an embodiment of the disclosure. FIG. **9** illustrates a top cover and a heat radiation duct, according to an embodiment of the disclosure.

[0165] As described above, the refrigerator **1** may include the heat radiation duct **700** arranged at the upper wall **110** to draw in air from outside of the main body to exchange heat with the heat radiation sink **520** and discharge the air that has warmed after the heat exchange with the heat radiation sink **520** back to the outside of the main body.

[0166] The heat radiation duct **700** may include the outside air intake port **751** arranged to draw in air from outside the main body, and outside air discharge ports **782** and **794** through which to discharge the air that has warmed after exchanging heat with the heat radiation sink **520** to the outside of the main body. The outside air intake port **751** may be formed on the top of the heat radiation duct **700**. At the outside air intake port **751**, a filter **390** may be arranged to prevent foreign materials from coming into the heat radiation duct **700** through the outside air intake port **751**. The filter **390** may be detachably arranged at the outside air intake port **751**.

[0167] The heat radiation duct **700** may include a fan receiver **760** for accommodating the heat radiation fan **600**, and a sink receiver **770** for accommodating the heat radiation sink **520**. The heat radiation duct **700** may include an intake duct part **750** arranged upstream of the fan receiver **760** to guide the air drawn in through the outside air intake port **751** to the fan receiver **760**. The heat radiation duct **700** may include discharge duct parts **780** and **790** arranged downstream of the sink receiver **770** to guide air that has warmed after exchanging heat with the heat radiation sink **520**.

[0168] The discharge duct parts **780** and **790** may include a first discharge duct part **780** for guiding the air that has warmed after exchanging heat with the heat radiation sink **520** to be discharged out of the main body **100**, and a second discharge duct part **790** branched from the first discharge duct part **780** for guiding the air that has warmed after exchanging heat with the heat radiation sink **520** to be discharged toward the rotation bar **40**. The first outside air discharge port **782** may be formed at the first discharge duct part **780**, and the second outside air discharge port **794** may be formed at the second discharge duct part **790**.

[0169] A fan receiving space **762** may be formed on the bottom of the fan receiver **760** to

accommodate the heat radiation fan **600**. A sink receiving space **771** may be formed on the bottom of the sink receiver **770** to accommodate the heat radiation sink **520**. An intake space **752** may be formed in the intake duct part **750** to guide the air drawn in through the outside air intake port **751** to the fan receiving space **762**.

[0170] When air is brought in from outside of the main body **100** through the outside air intake port **751**, the filter **390** arranged at the external intake port **751** may filter off foreign materials. The filter may be clogged by the foreign materials while filtering the foreign materials. The clogging of the filter **390** may include preventing the filtering due to the foreign materials. Furthermore, the clogging of the filter **390** may include a decrease in filtering efficiency due to the foreign materials.

[0171] In the case that the filter **390** is clogged, when the amount of air moving in the heat radiation duct **700** becomes smaller and air resistance decreases, the rotation speed of the heat radiation fan **600** operated at certain power may increase more than in the case that the filter **390** is not clogged.

[0172] When the filter **390** is clogged, causing air not to be properly brought in from outside of the main body **100**, the air brought in from outside of the main body **100** may not be efficiently discharged through the discharge duct parts **780** and **790** even when the heat radiation fan **600** is rotated. This may lead to a failure to normally lower the heat produced from the heater **531** and thus to a decrease in operation efficiency of the thermoelectric element **530**.

[0173] Hence, the filter **390** needs to be periodically cleaned or replaced when the filter **390** is clogged, but ordinary users are often unable to know of the cleaning or replacement cycle of the filter **390**. There is a need to notify whether the filter **390** is clogged to induce the user to clean or replace the filter **390** to increase the operation efficiency of the thermoelectric element **530**. For this, an embodiment according to the disclosure for determining and notifying whether the filter **390** is clogged will be described later in connection with FIGS. **10** to **18**.

[0174] A first discharge space **781** may be formed in the first discharge duct part **780** to guide air that has warmed after exchanging heat with the heat radiation sink **520** to the first outside air discharge port **782**. A second discharge space **791** may be formed in the second discharge duct part **790** to guide air that has warmed after exchanging heat with the heat radiation sink **520** to the second outside air discharge port **794**.

[0175] From another perspective, the heat radiation duct **700** may include a heat radiation duct body **710**, a heat radiation duct cover **720** and an extension duct **740**. In other words, the heat radiation duct body **710**, the heat radiation duct cover **720** and the extension duct **740** may be combined to form the heat radiation duct **700**. The heat radiation duct cover **720** may be coupled to the top of the heat radiation duct body **710**. The extension duct **740** may be arranged in front of the heat radiation duct body **710** to be arranged under the top cover **300**. The extension duct **740** may be arranged separately from or integrally with the heat radiation duct body **710**. The extension duct **740** may be coupled to a lower portion of the top cover **300**. The second outside air discharge port **794** may be formed between the extension duct **740** and the top cover **300**.

[0176] As described above, the refrigerator **1** may include a top cover **300** coupled to a top front portion of the main body **100** to cover the plurality of hinges **31**. The top cover **300** may include a top cover upper part **310**, a top cover side part **311** extending downward from edges of the top cover upper part **310**, and a top cover internal space **320** formed by the top cover upper part **310** and the top cover side part **311**.

[0177] The aforementioned filter **390** may be mounted at the top cover **300**. Specifically, the filter **390** may be mounted at the top cover **300** to be arranged on the outside air intake part **751** formed on the top of the heat radiation duct **700**. The top cover **300** may include an intake grill **350** formed on top of the outside air intake port **751**. Accordingly, coarse foreign materials may be primarily filtered off by the intake grill **350** from the air brought into the heat radiation duct **700** through the outside air intake port **751**, and fine foreign materials may be secondarily filtered off by the filter **390**.

[0178] The top cover **300** may include an extension duct coupler **380** coupled to the extension duct **740**. The top cover **300** may include a discharge port former **312** arranged on the front of the top cover **300** to form the second outside air discharge port **794** along with the extension duct **740**.

[0179] At least a portion of the air discharged from the heat radiation duct **700** through the first outside air discharge port **782** may flow into the top cover internal space **320**. In other words, the air that has warmed after exchanging heat with the heat radiation sink **520** may flow into the top cover internal space **320**. For this, a top cover inlet **330** may be formed at the top cover **300**.

[0180] The first outside air discharge port **782** may include a circulation discharge port **784** connected to the top cover inlet **330** to guide the air of the heat radiation duct **700** into the top cover internal space **320**. The first outside air discharge port **782** may include an external discharge port **783** divided from the circulation discharge port **784** to discharge the air of the heat radiation duct **700** to the outside of the top cover **300**.

[0181] The warm air brought into the top cover internal space **320** may circulate in the top cover internal space **320** and may be discharged out of the top cover **300**. For this, the top cover **300** may include a top cover outlet **340**. The warm air circulating in the top cover internal space **320** may heat the top of the main body **100**. This may prevent condensation on the top surface of the main body **100**.

[0182] The top cover **300** may include a discharge guide **381** formed to guide the air discharged out of the heat radiation duct **700** through the external discharge port **783** of the first outside air discharge port **782**. The discharge guide **381** may include a slope to guide the air discharged through the external discharge port **783** to be smoothly discharged without being interfered by the top cover **300**.

[0183] With the structure of the heat radiation duct **700** and the top cover **300**, the refrigerator **1** may include a first heat radiation path **401** through which the air that has warmed after passing through the heat radiation sink **520** is discharged out of the main body **100**, and a second heat radiation path **402** branched from the first heat radiation path **401** to allow the air that has warmed after passing through the heat radiation sink **520** to be discharged toward the rotation bar **40**. As the warm air after exchanging heat with the heat radiation sink **520** is discharged toward the rotation bar **40** through the second heat radiation path **402**, condensation on the rotation bar **40** may be prevented.

[0184] The refrigerator **1** may include a circulation path **388**, through which the air discharged through the first heat radiation path **401** is brought into the top cover **300**, circulating in the top cover **300**, and discharged out of the top cover **300**. As the warm air after exchanging heat with the heat radiation sink **520** circulates in the top cover **300** through the circulation path **388**, condensation on the top of the main body **100** may be prevented.

[0185] FIG. **10** is a control block diagram of a refrigerator, according to an embodiment of the disclosure.

[0186] Referring to FIG. **10**, the refrigerator **1** may include the thermoelectric cooling device **400**, the first speed sensor **111**, the second speed sensor **112**, the first temperature sensor **113**, the second temperature sensor **114**, a proximity sensor **116**, a power supplier **280**, the user interface device **260**, a communication interface **270** and/or a controller **360**.

[0187] The thermoelectric cooling device **400** may include the thermoelectric element **530** and/or a fan **50**.

[0188] When supplied with power, the thermoelectric element **530** may allow heat exchange between the cooling sink and the heat radiation sink. For example, the thermoelectric element **530** may convert electric energy to thermal energy so that heat radiation action occurs in the heater **531** and heat absorption action occurs in the cooler **532**.

[0189] When the heat radiation occurs in the heater **531**, the air that has warmed by the heat radiation sink **520** that contacts the heater **531** may be discharged out of the main body **100** and the air that has cooled by the cooling sink **570** that contacts the cooler **532** may be supplied into the

first storeroom **11**.

[0190] The controller **360** may control the thermoelectric element **530**. The controlling of the thermoelectric element **530** may include controlling the thermoelectric element **530** to be turned on/off. The controlling of the thermoelectric element **530** may include controlling the power supplier **280** to supply power to the thermoelectric element **530**.

[0191] The power supplier **280** may supply power to the thermoelectric element **530**. The power supplier **280** supplying power to the thermoelectric element **530** may include applying a certain current and/or voltage to the thermoelectric element **530**.

[0192] When the certain current and/or voltage is applied to the thermoelectric element **530**, the heat radiation sink **520** may come into contact with the heater **531** to absorb heat from the heater **531** and emit the heat out of the main body **100**.

[0193] When the certain current and/or voltage is applied to the thermoelectric element **530**, the cooling sink **570** may take heat from the first storeroom **11** and transfer the heat to the cooler **532**, thereby cooling the first storeroom **11**.

[0194] The fan **50** may include the heat radiation fan **600** and/or the cooling fan **800**.

[0195] The heat radiation fan **600** may be rotated according to the power supplied from the power supplier **280**.

[0196] The heat radiation fan **600** may be rotated by a heat radiation fan motor that receives power from the power supplier **280**.

[0197] The heat radiation fan motor may include a BLDC motor, a PMSM motor and/or a DC motor.

[0198] The heat radiation fan **600** may rotate to draw in air from outside the main body **100** and guide the air to exchange heat with the heat radiation sink **520**, and discharge the air that has exchanged heat with the heat radiation sink **520** back to the outside of the main body **100**.

[0199] The controller **360** may control the heat radiation fan **600**. The controlling of the heat radiation fan **600** may include operating the heat radiation fan **600** to rotate the heat radiation fan **600**. The operating of the heat radiation fan **600** may include supplying power to the heat radiation fan motor to rotate the heat radiation fan **600**.

[0200] The controlling of the heat radiation fan **600** may include stopping the rotation of the heat radiation fan **600**. The stopping of the rotation of the heat radiation fan **600** may include stopping the rotation of the heat radiation fan **600** by not supplying power to the heat radiation fan motor.

[0201] In an embodiment, the controller **360** may control the heat radiation fan **600** at multiple heat radiation levels. The controlling of the heat radiation fan **600** at multiple heat radiation levels may include operating the heat radiation fan **600** at multiple heat radiation levels. The operating of the heat radiation fan **600** at multiple heat radiation levels may include operating the heat radiation fan motor at multiple heat radiation levels.

[0202] The multiple heat radiation levels may include the first heat radiation level, the second heat radiation level and/or the third heat radiation level. The first heat radiation level may include a first preset voltage. The second heat radiation level may include a second preset voltage, and the third heat radiation level may include a third preset voltage. The third preset voltage may be higher than the second preset voltage. The second preset voltage may be higher than the first preset voltage.

[0203] The operating of the heat radiation fan motor at multiple heat radiation levels may include operating the heat radiation fan motor at the first level, operating the heat radiation fan motor at the second level, and operating the heat radiation fan motor at the third level.

[0204] The operating of the heat radiation fan motor at the first level may include applying the first preset voltage to the heat radiation fan motor. The operating of the heat radiation fan motor at the second level may include applying the second preset voltage to the heat radiation fan motor. The operating of the heat radiation fan motor at the third level may include applying the third preset voltage to the heat radiation fan motor.

[0205] In the disclosure, the multiple heat radiation levels are not limited thereto, and the number

of heat radiation levels may vary in various embodiments. For example, the number of the multiple heat radiation levels may be more than or less than three.

[0206] As the air that has exchanged heat with the heat radiation sink **520** is moved with the operation of the heat radiation fan **600**, the heat radiation sink **520** may radiate heat fast. With the rapid heat radiation of the heat radiation sink **520**, heat radiation action in the heater **531** and heat absorption action in the cooler **532** may occur smoothly.

[0207] The cooling fan **800** may be rotated according to the power supplied from the power supplier **280**.

[0208] The cooling fan **800** may be rotated by a cooling fan motor that receives power from the power supplier **280**.

[0209] The cooling fan motor may include a BLDC motor, a PMSM motor and/or a DC motor.

[0210] As the cooling fan **800** is rotated and the air that has exchanged heat with the cooling sink **570** is moved around, the inside of the first storeroom **11** may be cooled fast. With the movement of the air that has exchanged heat with the cooling sink **570**, heat radiation action in the heater **531** and heat absorption action in the cooler **532** may occur smoothly.

[0211] The controller **360** may control the cooling fan **800**. The controlling of the cooling fan **800** may include operating the cooling fan **800** to rotate the cooling fan **800**. The operating of the cooling fan **800** may include supplying power to the cooling fan motor to rotate the cooling fan **800**.

[0212] The controlling of the cooling fan **800** may include stopping the rotation of the cooling fan **800**. The stopping of the rotation of the cooling fan **800** may include stopping the rotation of the cooling fan **800** by not supplying power to the cooling fan motor.

[0213] In an embodiment, the controller **360** may control the cooling fan **800** at multiple cooling levels. The controlling of the cooling fan **800** at multiple cooling levels may include operating the cooling fan **800** at multiple cooling levels. The operating of the cooling fan **800** at multiple cooling levels may include operating the cooling fan motor at multiple cooling levels.

[0214] The multiple cooling levels may include the first cooling level, the second cooling level and/or the third cooling level. The first cooling level may include a first preset voltage. The second cooling level may include a second preset voltage, and the third cooling level may include a third preset voltage. The third preset voltage may be higher than the second preset voltage. The second preset voltage may be higher than the first preset voltage.

[0215] The operating of the cooling fan motor at multiple cooling levels may include operating the cooling fan motor at the first level, operating the cooling fan motor at the second level, and operating the cooling fan motor at the third level.

[0216] The operating of the cooling fan motor at the first level may include applying the first preset voltage to the cooling fan motor. The operating of the cooling fan motor at the second level may include applying the second preset voltage to the cooling fan motor. The operating of the cooling fan motor at the third level may include applying the third preset voltage to the cooling fan motor.

[0217] In the disclosure, the multiple cooling levels are not limited thereto, and the number of cooling levels may vary in various embodiments. For example, the number of multiple cooling levels may be more than or less than three.

[0218] The first speed sensor **111** may detect rotation speed of the heat radiation fan **600**. The first speed sensor **111** may detect the rotation speed of the heat radiation fan **600** by counting the electric signal that occurs each time the heat radiation fan **600** is rotated. The first speed sensor **111** may send information about the rotation speed of the heat radiation fan **600** to the controller **360**.

[0219] The second speed sensor **112** may detect rotation speed of the cooling fan **800**. The second speed sensor **112** may detect the rotation speed of the cooling fan **800** by counting the electric signal that occurs each time the cooling fan **800** is rotated. The second speed sensor **112** may send information about the rotation speed of the cooling fan **800** to the controller **360**.

[0220] The first temperature sensor **113** may detect temperature of the heater **531**. The first

temperature sensor **113** may send information about the temperature of the heater **531** to the controller **360**.

[0221] The second temperature sensor **114** may detect temperature of the cooler **532**. The second temperature sensor **114** may send information about the temperature of the cooler **532** to the controller **360**.

[0222] The proximity sensor **116** may detect an object outside the refrigerator **1**. For example, the proximity sensor **116** may detect a location of an object (e.g., the user) within a certain range from the main body. The proximity sensor **116** may send information about the object outside the refrigerator **1** to the controller **360**.

[0223] The power supplier **280** may supply power to various components of the refrigerator **1**. For example, the power supplier **280** may supply power to the various components of the refrigerator **1** based on a signal to power on the refrigerator **1**. The signal to power on the refrigerator **1** may include a signal that occurs from the refrigerator connected to a commercial power source.

[0224] The power supplier **280** may supply power to the various components of the refrigerator **1** based on a control signal of the controller **360**. For example, the power supplier **280** may supply power to the heat radiation fan **600** based on an operation control signal for the heat radiation fan **600** generated by the controller **360**.

[0225] The user interface device **260** may allow the user to interact with the refrigerator **1**.

[0226] The user interface device **260** may include the output interface **261** and the input interface **262**.

[0227] The at least one output interface **261** may send various information relating to operation of the refrigerator **1** to the user by generating sensory information.

[0228] For example, the at least one output interface **261** may deliver information relating to the settings of the refrigerator **1** and operation of the refrigerator **1** to the user. The information relating to an operation of the refrigerator **1** may be output through a display, an indicator and/or voice. The at least one output interface **261** may include, for example, a liquid crystal display (LCD) panel, an indicator, a light emitting diode (LED) panel, a speaker, etc.

[0229] In an embodiment, the at least one output interface **261** may output sensory information (e.g., visual information, auditory information, etc.) relating to control of the refrigerator **1**.

[0230] The at least one input interface **262** may convert sensory information received from the user into an electric signal.

[0231] When the user interface device **260** includes a touch screen display, the touch screen display may correspond to an example of the output interface **261** and the input interface **262**.

[0232] The at least one input interface **262** may include an input device (e.g., a button, a knob, etc.) for receiving a user input to control an operation of the refrigerator **1**.

[0233] Each button may include a visual indicator (e.g., text, an icon, etc.) that may indicate the function.

[0234] The at least one input interface **262** may include, for example, a tact switch, a push switch, a slide switch, a toggle switch, a micro switch, a touch switch, a touch pad, a touch screen, a jog dial, and/or a microphone.

[0235] In the disclosure, the button may be replaced by a user interface (UI) element, a tact switch, a push switch, a slide switch, a toggle switch, a micro switch, a touch switch, a touch pad, a touch screen, a jog dial, and/or a microphone.

[0236] The refrigerator **1** may process the user input received through the user interface device **260** and output information relating to the refrigerator **1** through the user interface device **260**.

[0237] The refrigerator **1** may control the operation of the refrigerator **1** based on the user input received through the user interface device **260**.

[0238] The refrigerator **1** may include the communication interface **270** for communicating with an external device (e.g., a server **20** or a user device **30**, see FIG. **11**) wiredly and/or wirelessly.

[0239] The communication interface **270** may include at least one of a short-range communication

module or a long-range communication module.

[0240] The communication interface **270** may transmit data to the external device (e.g., the server **20** or the user device **30**, see FIG. **11**) or receive data from the external device. For this, the communication interface **270** may support establishment of a direct (e.g., wired) communication channel or a wireless communication channel between external devices (e.g., the server **20** or the user device **30**, see FIG. **11**), and communication through the established communication channel. In an embodiment, the communication interface **270** may include a wireless communication module (e.g., a cellular communication module, a short-range wireless communication module or a GNSS communication module) or a wired communication module (e.g., a LAN communication module or a power-line communication module). A corresponding one of the communication modules may communicate with the external device (e.g., the server **20** or the user device **30**, see FIG. **11**) over a first network (e.g., a short-range communication network such as bluetooth, wireless-fidelity (Wi-Fi) direct or infrared data association (IrDA)) or a second network (e.g., a remote communication network such as a legacy cellular network, a fifth generation (5G) network, a next generation communication network, the Internet, or a computer network (e.g., a LAN or wide area network (WAN))). These various types of communication modules may be integrated into a single component (e.g., a single chip) or implemented as a plurality of separate components (e.g., a plurality of chips).

[0241] The short-range communication module may include a bluetooth communication module, a BLE communication module, an NFC module, a WLAN, e.g., Wi-Fi, communication module, a Zigbee communication module, an IrDA communication module, a WFD communication module, an UWB communication module, an Ant+ communication module, a uWave communication module, etc., without being limited thereto.

[0242] The long-range communication module may include a communication module for performing various types of long-range communication and include a mobile communication interface **270**. The mobile communication interface **270** transmits or receives wireless signals to and from at least one of a base station, an external terminal, or a server in a mobile communication network.

[0243] In an embodiment, the communication interface **270** may communicate with the external device through a nearby access point (AP). The AP may connect a local area network (LAN) connected to the refrigerator **1** to a wide area network (WAN) connected to the server. The refrigerator **1** may be connected to the server through the WAN.

[0244] The refrigerator **1** may receive various signals (e.g., weather information or remote command) from the external device (e.g., the server **20** or the user device **30**, see FIG. **11**) through the communication interface **270**.

[0245] The refrigerator **1** may transmit various signals to the external device (e.g., the server **20** or the user device **30**, see FIG. **11**) through the communication interface **270**.

[0246] Various information received by the refrigerator **1** from the external device (e.g., the server **20** or the user device **30**, see FIG. **11**) through the communication interface **270** and various information transmitted by the refrigerator **1** to the external device (e.g., the server **20** or the user device **30**) through the communication interface **270** will be described later with reference to FIG. **11**.

[0247] The controller **360** may include at least one processor **361** for controlling operation of the refrigerator **1**, and at least one memory **362** for storing a program and data for controlling the operation of the refrigerator **1**.

[0248] The at least one memory **362** may store data required for various embodiments. The memory **362** may be implemented in the form of a memory **362** embedded in or detachable from the refrigerator **1** depending on the data storage use. For example, data for operating the refrigerator **1** may be stored in the memory **362** embedded in the refrigerator **1** and data for an extended function of the refrigerator **1** may be stored in the memory detachable from the

refrigerator **1**. In the meantime, the memory **362** embedded in the refrigerator **1** may be implemented with at least one of a volatile memory (e.g., a dynamic random access memory (DRAM), a static RAM (SRAM), or a synchronous dynamic RAM (SDRAM), etc.) or the non-volatile memory **362** (e.g., a one time programmable read only memory (OTPROM), a programmable ROM (PROM), an erasable and programmable ROM (EPROM), an electrically erasable and programmable ROM (EEPROM), a mask ROM, a flash ROM, a flash memory **362** (e.g., NAND flash or NOR flash), a hard drive or a solid state drive (SSD)). The memory detachable from the refrigerator **1** may be implemented in such a format as a memory card (e.g., compact flash (CF), secure digital (SD), micro-SD, mini-SD, extreme digital (xD), multi-media card (MMC), etc.) or an external memory (e.g., USB memory) connectable to a USB port.

[0249] The at least one processor **361** controls general operation of the refrigerator **1**. Specifically, the at least one processor **361** may be connected to the respective components (e.g., the thermoelectric cooling device **400**, the first speed sensor **111**, the second speed sensor **112**, the first temperature sensor **113**, the second temperature sensor **114**, the power supplier **280**, the user interface device **260** and/or the communication interface **270**) of the refrigerator **1** to control general operation of the refrigerator **1**. For example, the at least one processor **361** may be electrically connected to the memory **362** to control general operation of the refrigerator **1**. The processor **361** may be configured with one or more processors.

[0250] The at least one processor **361** may execute at least one instruction stored in the memory **362** to perform operation of the refrigerator **1** according to various embodiments.

[0251] The at least one processor **361** may include one or more of a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), a many integrated core (MIC), a digital signal processor (DSP), a neural processing unit (NPU), a hardware accelerator or a machine learning accelerator. The at least one processor **361** may control one or any combination of the other components of the refrigerator **1**, and perform a communication related operation or data processing. The at least one processor **361** may execute at least one program or instruction stored in the memory **362**. For example, the at least one processor **361** may execute the at least one instruction stored in the memory **362** to perform a method according to at least one embodiment of the disclosure.

[0252] FIG. **11** is a conceptual diagram illustrating an example of an entity to perform a method of controlling the refrigerator **1**, according to an embodiment of the disclosure.

[0253] In an embodiment, the communicating of the refrigerator **1** with the user device **30** may include establishing direct connection of communication between the refrigerator **1** and the user device **30** and/or establishing indirect connection of communication between the refrigerator **1** and the user device **30** via the server **20**.

[0254] The establishing of the direct connection of communication between the refrigerator **1** and the user device **30** may refer to the refrigerator **1** transmitting various signals directly to the user device **30** and the user device **30** transmitting various signals directly to the refrigerator **1**.

[0255] The establishing of indirect connection of communication between the refrigerator **1** and the user device **30** through the server **20** may refer to an occasion when the refrigerator **1** transmits various signals to the server **20**, which in turn forwards them to the user device **30**, and the user device **30** transmits various signals to the server **20**, which in turn forwards them to the refrigerator **1**.

[0256] In the disclosure, the various signals may include information, data, control commands, etc.

[0257] In the disclosure, the user device **30** may refer to an electronic device capable of wireless communication, such as a smartphone, a personal computer (PC), etc.

[0258] In the disclosure, the server **20** may refer to a computing device capable of wireless communication.

[0259] Each of the server **20** and the user device **30** may also include a communication module for communicating with the refrigerator **1**.

[0260] FIG. 12 illustrates an example of a flowchart of a method of controlling the refrigerator 1, according to an embodiment of the disclosure.

[0261] Referring to FIG. 12, the controller 360 may operate the fan 50, in 1000. The controller 360 may operate the heat radiation fan 600 and/or the cooling fan 800.

[0262] The rotation speed of the fan 50 may vary depending on the state of the filter 390. For example, referring to FIG. 8, when the filter 390 for filtering foreign materials of the air brought in from outside of the main body 100 is clogged, the inflow rate of air circulating in the heat radiation duct 700 may decrease, which leads to a decrease in air resistance in the heat radiation duct 700 and thus, to an increase in rotation speed of the heat radiation fan 600. In another example, although not shown, when the filter 390 is arranged at the inside air intake port 991 (see FIG. 3) in order to filter off the foreign materials of the air inside the storeroom 11 through the inside air intake port 992, the inflow rate of the air circulating in the cooling duct 900 may decrease, which leads to a decrease in air resistance in the cooling duct 900 and to an increase in rotation speed of the cooling fan 800.

[0263] The controller 360 may obtain rotation speed of the heat radiation fan 600. For example, the controller 360 may obtain the rotation speed of the heat radiation fan 600 by receiving information about the rotation speed of the heat radiation fan 600 detected by the first speed sensor 111.

[0264] The controller 360 may obtain rotation speed of the cooling fan 800. For example, the controller 360 may obtain the rotation speed of the cooling fan 800 by receiving information about the rotation speed of the cooling fan 800 detected by the second speed sensor 112.

[0265] In various embodiments, the controller 360 may notify clogging of the filter 390 in response to the rotation speed of the fan 50 being higher than a reference rotation speed.

[0266] In an embodiment, the controller 360 may notify clogging of the filter 390 in response to the rotation speed of the heat radiation fan 600 being higher than the reference rotation speed.

[0267] For example, the controller 360 may apply a preset voltage to the heat radiation fan motor, determine that the filter 390 is clogged when the rotation speed of the heat radiation fan 600 is higher than the reference rotation speed, and notify the clogging of the filter 390.

[0268] When the filter 390 is located in the cooling duct 900, in an embodiment, the controller 360 may notify clogging of the filter 390 in response to the rotation speed of the cooling fan 800 being higher than a reference rotation speed.

[0269] For example, the controller 360 may apply a preset voltage to the motor for the cooling fan 800, determine that the filter 390 is clogged when the rotation speed of the cooling fan 800 is higher than the reference rotation speed, and notify the clogging of the filter 390.

[0270] In other words, when the filter 390 is arranged at the outside air intake port 751 as shown in FIG. 8, the controller 360 may notify clogging of the filter 390 in response to the rotation speed of the heat radiation fan 600 being higher than the reference speed. On the other hand, although not shown, when the filter 390 is located at the inside air intake port 991 (see FIG. 3), the controller 360 may notify clogging of the filter 390 in response to the rotation speed of the cooling fan 800 being higher than the reference speed.

[0271] For convenience of explanation, the fan 50 will now be described as referring to the heat radiation fan 600.

[0272] FIG. 13 illustrates an example of an interface provided by the refrigerator 1 or a user device, according to an embodiment of the disclosure.

[0273] Referring to FIG. 13, in various embodiments, the controller 360 may notify clogging of the filter 390 in response to the rotation speed of the fan 50 being higher than a reference rotation speed.

[0274] In an embodiment, the controller 360 may notify clogging of the filter 390 through the user interface device 260.

[0275] For example, the controller 360 may display a first interface U1 on the display to indicate that the filter 390 needs to be cleaned due to clogging of the filter 390 in response to the rotation

speed of the fan **50** being higher than the reference rotation speed.

[0276] The first interface **U1** may include a visual indicator **U11** indicating that the filter **390** needs to be cleaned, and/or an interface element **U12** for receiving a user input indicating the start of cleaning of the filter **390**.

[0277] The user input indicating the start of cleaning of the filter **390** may include an input from the user through the first interface **U1** when the user plans to replace or clean the filter **390**.

[0278] In another example, the controller **360** may output a guidance voice indicating that the filter **390** needs to be cleaned through a speaker in response to the rotation speed of the fan **50** being higher than the reference rotation speed.

[0279] In an embodiment, the controller **360** may notify clogging of the filter **390** at predetermined intervals.

[0280] For example, after notifying the clogging of the filter **390** in response to the rotation speed of the fan **50** being higher than the reference rotation speed, the controller **360** may display the first interface **U1** on the display or output guidance voice indicating that the filter **390** needs to be cleaned through a speaker at the predetermined intervals (e.g., every 24 hours) based on a failure to receive the user input indicating the start of cleaning of the filter **390** for a predetermined period of time (e.g., for 10 minutes).

[0281] In an embodiment, the controller **360** may notify clogging of the filter **390** based on an object outside the refrigerator **1** being detected.

[0282] For example, after notifying the clogging of the filter **390** in response to the rotation speed of the fan **50** being higher than the reference rotation speed, the controller **360** may display the first interface **U1** on the display or output guidance voice indicating that the filter **390** needs to be cleaned through a speaker based on an object (e.g., the user) being detected within a certain distance from the main body **100** through the proximity sensor **116**.

[0283] In an embodiment, the controller **360** may transmit information regarding the clogging of the filter **390** to the server **20** and/or the user device **30** through the communication interface **270** in response to the rotation speed of the fan **50** being higher than the reference rotation speed. The information regarding the clogging of the filter **390** may include whether the filter **390** is clogged, the rotation speed of the fan **50** and/or the reference rotation speed. The user device may notify the clogging of the filter **390**.

[0284] For example, the controller **360** may transmit information regarding whether the filter **390** is clogged to the server through the communication interface **270** in response to the rotation speed of the fan **50** being higher than the reference rotation speed; the server may forward the information regarding whether the filter **390** is clogged to the user device **30**; the user device **30** may display the first interface **U1**.

[0285] In another example, the controller **360** may transmit information about the rotation speed of the fan **50** and information about the reference rotation speed to the server **20** through the communication interface **270**, and the server may determine whether the filter **390** is clogged in response to the rotation speed of the fan **50** being higher than the reference rotation speed and transmit information regarding whether the filter **390** is clogged to the user device **30**.

[0286] In another example, the controller **360** may transmit information regarding whether the filter **390** is clogged to the user device **30** through the communication interface **270** in response to the rotation speed of the fan **50** being higher than the reference rotation speed, and the user device **30** may display the first interface **U1**.

[0287] FIG. **14** illustrates another example of an interface provided by the refrigerator **1** or the user device **30**, which is different from FIG. **13**, according to an embodiment of the disclosure.

[0288] Referring to FIG. **14**, the controller **360** may provide a second interface **U2** for setting an operation of the refrigerator **1** through the output interface **261**. For example, the controller **360** may display the second interface **U2** through the display.

[0289] In an embodiment, the controller **360** may provide a third interface **U3** through the output

interface **261** for changing the settings for cleaning of the filter **390** based on a user input to change the settings for an operation of the refrigerator **1** being received through the second interface **U2** (see (a) of FIG. **14** and (b) of FIG. **14**).

[0290] For example, the controller **360** may display the third interface **U3** through the display based on a user input to change the settings for an operation of the refrigerator **1** being received through the second interface **U2** (see (a) of FIG. **14** and (b) of FIG. **14**).

[0291] Specifically, the controller **360** may receive the user input to change settings for an operation of the refrigerator **1** through the second interface **U2**, and display the third interface **U3** including a visual indicator **U31** indicating that the filter **390** needs to be cleaned through the display in response to the rotation speed of the fan **50** being higher than the reference rotation speed.

[0292] Furthermore, the controller **360** may receive the user input to change settings for an operation of the refrigerator **1** through the second interface **U2**, and output guidance voice indicating that the filter **390** needs to be cleaned through a speaker in response to the rotation speed of the fan **50** being higher than the reference rotation speed.

[0293] Specifically, the controller **360** may display the third interface **U3** including the visual indicator **U31** indicating that the filter **390** needs to be cleaned through the display (see (b) of FIG. **14**) or output guidance voice indicating that the filter **390** needs to be cleaned through a speaker only when the rotation speed of the fan **50** is higher than the reference rotation speed, and may display the third interface **U3** that does not include the visual indicator **U31** that indicates that the filter **390** needs to be cleaned through the display (see (a) of FIG. **14**) when the rotation speed of the fan **50** is equal to or lower than the reference rotation speed.

[0294] The user device **30** may display the second interface **U2** for setting an operation of the refrigerator **1**.

[0295] The user device **30** may display the third interface **U3** based on a user input to change the settings for an operation of the refrigerator **1** being received through the second interface **U2**.

[0296] For example, the user device **30** may receive a user input to change settings for an operation of the refrigerator **1** through the second interface **U2**, and display the third interface **U3** including a visual indicator **U31** indicating that the filter **390** needs to be cleaned based on the information regarding whether the filter **390** is clogged being received from the refrigerator **1** and/or the server **20**.

[0297] FIG. **15** illustrates an example of an interface provided by the refrigerator **1** or the user device when a user input is received through the interface of FIG. **13** or **14**.

[0298] Referring to FIG. **15**, the controller **360** may display a fourth interface **U4** including information about a method of cleaning the filter **390** through the display based on a user input indicating the start of cleaning of the filter **390** being received through the first interface **U1** (see FIG. **13**).

[0299] Furthermore, the controller **360** may display the fourth interface **U4** including information about a method of cleaning the filter **390** through the display based on a user input indicating the start of cleaning of the filter **390** being received through the third interface **U3** (see FIG. **14**).

[0300] The fourth interface **U4** may sequentially include the information about the method of cleaning the filter **390**. For example, the fourth interface **U4** may provide a visual indicator that may notify the information about the method of cleaning the filter **390** to the user in stages (see (a), (b) and (c) of FIG. **15**).

[0301] After displaying the fourth interface **U4** through the display, the controller **360** may display a fifth interface **U5** for receiving a user input that indicates completion of cleaning of the filter **390** through the display.

[0302] The fifth interface **U5** may include an interface element **U51** for receiving the user input that indicates completion of cleaning of the filter **390**.

[0303] The controller **360** may display a sixth interface **U6** indicating that cleaning of the filter is

completed through the display or output guidance voice indicating that cleaning of the filter is completed through a speaker, based on the user input indicating that cleaning of the filter **390** is completed being received through the interface element **U51**.

[0304] Even when the user input indicating that cleaning of the filter is completed is received from the user, cleaning or replacement of the filter may not properly performed sometimes. Hence, there is a need to induce the user to clean or replace the filter by notifying the clogging of the filter again when the cleaning or replacement of the filter is not properly performed, which will now be described in an embodiment of the disclosure.

[0305] FIG. **16** illustrates an example of a flowchart of a method of controlling the refrigerator **1**, according to an embodiment of the disclosure.

[0306] Referring to FIG. **16**, the controller **360** may power on the refrigerator **1** through the power supplier **280**, in **1100**. The powering on of the refrigerator **1** may refer to an occasion when an operation of the refrigerator **1** connected to a commercial power source is started and power is supplied to various components of the refrigerator **1** through the power supplier **280**.

[0307] In an embodiment, based on a lapse of a first preset period of time after the refrigerator **1** is powered on, the controller **360** may operate the fan **50** for a second preset period of time.

[0308] For example, after a lapse of the first preset period of time (e.g., one hour) in **1100** of FIG. **16** after the refrigerator **1** is powered on, the controller **360** may operate the fan for the second preset period of time (e.g., one minute) in **1120** of FIG. **16**.

[0309] In an embodiment, the controller **360** may set the reference rotation speed in **1130**.

[0310] For example, the controller **360** may obtain rotation speed of the heat radiation fan **600** detected by the first speed sensor **111** for the second preset period of time, and set a value larger than the rotation speed of the heat radiation fan **600** detected for the second preset period of time by a predetermined value to the reference rotation speed.

[0311] In the disclosure, the setting of the reference rotation speed may include setting the reference rotation speed according to the rotation speed of the fan **50** for the second preset period of time after a lapse of the first preset period of time after the refrigerator **1** is powered on, but the reference rotation speed may be set according to many different embodiments.

[0312] For example, the reference rotation speed may be set to be a preset rotation speed before the refrigerator **1** is installed at home, and the rotation speed of the fan **50** obtained right after the refrigerator **1** is powered on may be set as the reference rotation speed.

[0313] However, rotation speed of the fan **50** may vary according to the installation environment of the refrigerator **1**, and considering time for system stabilization after the refrigerator **1** is powered on, the reference rotation speed will be described as being set based on the rotation speed of the fan **50** obtained for the second preset period of time after a lapse of the first preset period of time after the refrigerator **1** is powered on.

[0314] In an embodiment, in response to the rotation speed of the fan **50** being higher than the reference rotation speed in **1140**, the controller **360** may notify clogging of the filter **390** in **1150**.

[0315] For example, the controller **360** may notify the clogging of the filter **390** in response to the rotation speed of the fan **50** obtained in the second preset period of time being higher than the reference rotation speed set based on the rotation speed of the fan **50** obtained for the second preset period of time.

[0316] In an embodiment, the controller **360** may re-notify the clogging of the filter **390** when no user input indicating that cleaning of the filter **390** is completed is received, in **1150**.

[0317] For example, when no user input indicating that cleaning of the filter **390** is completed after clogging of the filter **390** is notified, the controller **360** may re-notify the clogging of the filter **390** at preset intervals (e.g., every 24 hours).

[0318] In an embodiment, based on the user input indicating completion of cleaning of the filter **390** being received in **1160** after the clogging of the filter **390** is notified, the controller **360** may operate the fan **50** for a third preset period of time in **1170**.

[0319] The controller **360** may determine whether the cleaning of the filter **390** is completed.

[0320] In an embodiment, the controller **360** may determine whether the cleaning of the filter **390** is completed based on a difference between the rotation speed of the fan **50** for the third preset period of time and the reference rotation speed.

[0321] For example, when the difference between the rotation speed of the fan **50** for the third preset period of time and the reference rotation speed being larger than a preset difference after the fan **50** is operated for the third preset time based on the user input indicating completion of cleaning of the filter **390** being received after the clogging of the filter **390** is notified, the controller **360** may determine that the cleaning of the filter **390** is not completed.

[0322] In another example, when the difference between the rotation speed of the fan **50** for the third preset period of time and the reference rotation speed being smaller than the preset difference after the fan **50** is operated for the third preset time based on the user input indicating completion of cleaning of the filter **390** being received after the clogging of the filter **390** is notified, the controller **360** may determine that the cleaning of the filter **390** is completed.

[0323] In an embodiment, when the cleaning of the filter **390** is determined as being completed in **1180**, the controller **360** may reset the reference rotation speed in **1190**.

[0324] In an embodiment, the controller **360** may re-notify clogging of the filter **390** in **1180** when it is determined that cleaning of the filter **390** is not completed.

[0325] In an embodiment, the controller **360** may notify clogging of the filter **390** in response to the rotation speed of the fan **50** being higher than the reset reference rotation speed.

[0326] For example, when it is determined that the cleaning of the filter **390** is completed and the reference rotation speed is reset, the controller **360** may notify clogging of the filter **390** in response to the rotation speed of the fan **50** being higher than the reset reference rotation speed.

[0327] In the disclosure, after the refrigerator **1** is powered on, in response to the rotation speed of the fan **50** being higher than the reference rotation speed, clogging of the filter **390** is notified, and the reference rotation speed is reset when it is determined that the cleaning of the filter **390** is completed. Specifically, the reference rotation speed may be reset based on the rotation speed of the fan **50** being higher than the reference rotation speed after clogging of the filter **390** is notified each time the clogging of the filter **390** is notified.

[0328] In the disclosure, the filter **390** or the fan **50** may have varying performance over time. Accordingly, it is more advantageous in that whether the filter **390** is clogged may be more accurately determined by updating the reference rotation speed for determining whether the filter **390** is clogged based on the rotation speed of the fan **50** each time cleaning of the filter **3** is completed.

[0329] FIG. **17** illustrates an example of a flowchart of a method of controlling a refrigerator, according to an embodiment of the disclosure.

[0330] Referring to FIG. **17**, based on a lapse of the first preset period of time in **1210** after the refrigerator **1** is powered on in **1200**, the controller **360** may operate the fan **50** for the second preset period of time at multiple levels.

[0331] In an embodiment, the controller **360** may operate the fan **50** for the second preset period of time at multiple levels, in **1220**.

[0332] For example, based on a lapse of the first preset period of time in **1210** after being powered on in **1200**, the controller **360** may operate the fan **50** for the second preset period of time at multiple levels in **1230**.

[0333] The operating of the fan **50** by the controller **360** for the second preset period of time at multiple levels may include operating the fan **50** at the first level.

[0334] The operating of the fan **50** by the controller **360** for the second preset period of time at multiple levels may include operating the fan **50** at the second level.

[0335] The operating of the fan **50** by the controller **360** for the second preset period of time at multiple levels may include operating the fan **50** at the third level.

[0336] The controller **360** may operate the fan **50** at multiple levels at preset intervals (e.g., every one minute) during the second preset period of time.

[0337] The multiple levels were described above, so the description will not be repeated.

[0338] For example, after operating the fan **50** for a preset period of time at the first level during the second preset period of time, the controller **360** may operate the fan **50** for the preset period of time at the second level and then operate the fan **50** for the preset period of time at the third level.

[0339] In an embodiment, the controller **360** may set a plurality of reference rotation speeds in **1230**.

[0340] For example, the controller **360** may set a first reference rotation speed based on the rotation speed of the fan **50** obtained by operating the fan **50** for the preset period of time at the first level during the second preset period of time.

[0341] In another example, after operating the fan **50** for the preset period of time at the first level during the second preset period of time, the controller **360** may set the second reference rotation speed based on the rotation speed of the fan **50** obtained by operating the fan **50** for the preset period of time at the second level.

[0342] In another example, after operating the fan **50** for the preset period of time at the second level during the second preset period of time, the controller **360** may set the third reference rotation speed based on the rotation speed of the fan **50** obtained by operating the fan **50** for the preset period of time at the third level.

[0343] In the disclosure, the number of reference rotation speeds is not limited thereto, and may be set depending on various embodiments.

[0344] In various embodiments, the controller **360** may notify clogging of the filter in **1250**, based on the rotation speed of the fan **50** operated at one of multiple levels being higher than the reference rotation speed set by operating at corresponding levels in **1240**. In this case, the notifying of the clogging of the filter is equal to what is described above.

[0345] In an embodiment, the controller **360** may notify clogging of the filter **390** in response to the rotation speed of the fan **50** being higher than the first reference rotation speed when the fan **50** is operated at the first level after the first reference rotation speed is set.

[0346] In an embodiment, the controller **360** may notify clogging of the filter **390** in response to the rotation speed of the fan **50** being higher than the second reference rotation speed when the fan **50** is operated at the second level after the second reference rotation speed is set.

[0347] In an embodiment, based on the user input indicating completion of cleaning of the filter **390** being received after the clogging of the filter **390** is notified, the controller **360** may operate the fan **50** for the third preset period of time at multiple levels in **1270**.

[0348] For example, based on the user input indicating completion of cleaning of the filter **390** being received after the clogging of the filter **390** is notified, the controller **360** may operate the fan **50** for the third preset period of time at the first level.

[0349] In another example, based on the user input indicating completion of cleaning of the filter **390** being received after the clogging of the filter **390** is notified, the controller **360** may operate the fan **50** at the first level and then operate the fan **50** at the second level during the third preset period of time.

[0350] In another example, based on the user input indicating completion of cleaning of the filter **390** being received after the clogging of the filter **390** is notified, the controller **360** may operate the fan **50** at the second level and then operate the fan **50** at the third level during the third preset period of time.

[0351] In an embodiment, the controller **360** may determine whether the cleaning of the filter **390** is completed based on a difference between the rotation speed of the fan **50** obtained by operating the fan **50** at multiple levels for the third preset period of time and the reference rotation speed set by operating at corresponding levels.

[0352] For example, the controller **360** may determine that the cleaning of the filter **390** is

completed based on the difference between the rotation speed of the fan **50** obtained by operating the fan **50** at the first level for the third preset period of time and the corresponding first reference rotation speed being equal to or less than a reference difference.

[0353] In another embodiment, the controller **360** may determine that the cleaning of the filter **390** is completed based on a difference between the rotation speed of the fan **50** obtained by operating the fan **50** at the second level for the third preset period of time and the corresponding second reference rotation speed being equal to or less than the reference difference.

[0354] In another embodiment, the controller **360** may determine that the cleaning of the filter **390** is completed based on a difference between the rotation speed of the fan **50** obtained by operating the fan **50** at the third level for the third preset period of time and the corresponding third reference rotation speed being equal to or less than the reference difference.

[0355] In an embodiment, the controller **360** may reset the reference rotation speed when the cleaning of the filter **390** is determined as being completed.

[0356] For example, the controller **360** may reset the reference rotation speed when the cleaning of the filter **390** is determined as being completed.

[0357] Specifically, based on a difference between the rotation speed of the fan **50** and the first reference rotation speed when the fan **50** is operated at the first level during the third preset period of time, the controller **360** may determine whether the cleaning of the filter **390** is completed, and reset the first reference rotation speed when it is determined that cleaning of the filter **390** is completed.

[0358] FIG. **18** illustrates an example of a flowchart of a method of controlling a refrigerator for notifying filter clogging based on rotation speed of a fan and temperature of a thermoelectric element, according to an embodiment of the disclosure.

[0359] Referring to FIG. **18**, the controller **360** may operate the thermoelectric element **530**, in **2000**.

[0360] In an embodiment, in response to the rotation speed of the fan **50** being higher than the reference rotation speed in **2100** and a temperature drop rate of at least one of the heater **531** and the cooler **532** is smaller than a reference value in **2200** after the thermoelectric element **530** is operated, the controller **360** may notify clogging of the filter **390** in **2300**.

[0361] For example, when the filter **390** is clogged, operation efficiency of the thermoelectric element **530** may be lowered because the air in the heat radiation duct **700** is not properly circulated. As a result, the rate of temperature change of the heater **531** and/or the cooler **532** may be lower than in a normal state during the operation of the thermoelectric element **530**, so the controller **360** may notify the clogging of the filter **390** based on the rotation speed of the fan **50** being higher than the reference rotation speed and the rate of temperature change of at least one of the heater **531** and the cooler **532** being lower than a reference value.

[0362] According to an embodiment of the disclosure, a refrigerator may include a main body forming a storeroom; a thermoelectric element including a heater and a cooler; a fan configured to blow air heated by the heater out of the main body or blow air cooled by the cooler into the storeroom; a filter configured to filter off foreign materials in air brought in from outside of the main body by operation of the fan; and a controller configured to notify clogging of the filter in response to rotation speed of the fan being higher than a reference rotation speed.

[0363] The controller may be configured to, based on a lapse of a first preset period of time after the refrigerator is powered on, operate the fan for a second preset period of time, and based on the rotation speed of the fan during the second preset period of time, set the reference rotation speed.

[0364] The controller may operate the fan for a third preset period of time based on a user input indicating completion of filter cleaning being received after the clogging of the filter is notified.

[0365] The controller may determine whether cleaning of the filter is completed based on a difference between the rotation speed of the fan during the third preset period of time and the reference rotation speed.

[0366] The controller may reset the reference rotation speed in response to determining of completion of cleaning of the filter.

[0367] The controller may re-notify clogging of the filter in response to determining that cleaning of the filter is not completed.

[0368] The reference rotation speed may include a first reference rotation speed and a second reference rotation speed, and the controller may be configured to set the first reference rotation speed by operating the fan at a first level for the second preset period of time, set the second reference rotation speed by operating the fan at a second level higher than the first level for the second preset period of time, notify clogging of the filter **390** in response to the rotation speed of the fan operated at the first level being higher than the first reference rotation speed, and notify clogging of the filter **390** in response to the rotation speed of the fan operated at the second level being higher than the second reference rotation speed.

[0369] Based on a user input indicating that cleaning of the filter is completed being received after clogging of the filter is notified, the controller may be configured to determine whether cleaning of the filter is completed based on a difference between the rotation speed of the fan operated at the first level for the third preset period of time and the first reference rotation speed, and determine whether cleaning of the filter is completed based on a difference between the rotation speed of the fan operated at the second level for the third preset period of time and the second reference rotation speed.

[0370] The controller may be configured to notify clogging of the filter in response to the rotation speed of the fan being higher than the reference rotation speed and a temperature drop rate of at least one of the heater and the cooler being smaller than a reference value after the thermoelectric element is operated.

[0371] A communication interface for communicating with an external device may be further included, and the controller may be configured to transmit information regarding the clogging of the filter to the external device through the communication interface in response to the rotation speed of the fan being higher than the reference rotation speed.

[0372] According to an embodiment of the disclosure, a method of controlling a refrigerator including a main body forming a storeroom, a thermoelectric element including a heater and a cooler, a fan for blowing air heated by the heater to outside of the main body or blowing air cooled by the cooler into the storeroom, and a filter for filtering off foreign materials in air brought in from outside of the main body by operation of the fan may include notifying clogging of the filter in response to rotation speed of the fan being higher than a reference rotation speed.

[0373] The method of controlling the refrigerator may further include, based on a lapse of a first preset period of time after the refrigerator is powered on, operating the fan for a second preset period of time, and based on the rotation speed of the fan during the second preset period of time, setting the reference rotation speed.

[0374] The method of controlling the refrigerator may further include operating the fan for a third preset period of time based on a user input indicating completion of filter cleaning being received after the clogging of the filter is notified.

[0375] The method of controlling the refrigerator may further include determining whether cleaning of the filter is completed based on a difference between the rotation speed of the fan during the third preset period of time and the reference rotation speed.

[0376] The method of controlling the refrigerator may further include resetting the reference rotation speed in response to determining that cleaning of the filter is completed.

[0377] The method of controlling the refrigerator may further include re-notifying clogging of the filter in response to determining that cleaning of the filter is not completed.

[0378] The reference rotation speed may include a first reference rotation speed and a second reference rotation speed, the method of controlling the refrigerator may further include setting the first reference rotation speed by operating the fan at a first level for the second preset period of

time, and setting the second reference rotation speed by operating the fan at a second level higher than the first level for the second preset period of time, and the notifying of the clogging of the filter may include notifying clogging of the filter in response to the rotation speed of the fan operated at the first level being higher than the first reference rotation speed, and notifying clogging of the filter in response to the rotation speed of the fan operated at the second level being higher than the second reference rotation speed.

[0379] Based on a user input indicating that cleaning of the filter is completed being received after clogging of the filter is notified, the method of controlling the refrigerator may further include determining whether cleaning of the filter is completed based on a difference between the rotation speed of the fan operated at the first level for the third preset period of time and the first reference rotation speed, and determining whether cleaning of the filter is completed based on a difference between the rotation speed of the fan operated at the second level for the third preset period of time and the second reference rotation speed.

[0380] The notifying of the clogging of the filter may include notifying clogging of the filter in response to the rotation speed of the fan being higher than the reference rotation speed and a temperature drop rate of at least one of the heater and the cooler being smaller than a reference value after the thermoelectric element is operated.

[0381] The method of controlling the refrigerator may further include notifying information regarding the clogging of the filter to an external device in response to rotation speed of the fan being higher than a reference rotation speed.

[0382] Meanwhile, the embodiments of the disclosure may be implemented in the form of a recording medium for storing instructions to be carried out by a computer. The instructions may be stored in the form of program codes, and when executed by a processor, may generate program modules to perform operations in the embodiments of the disclosure. The recording media may correspond to computer-readable recording media.

[0383] The computer-readable recording medium includes any type of recording medium having data stored thereon that may be thereafter read by a computer. For example, it may be a read only memory (ROM), a random access memory (RAM), a magnetic tape, a magnetic disk, a flash memory, an optical data storage device, etc.

[0384] The computer-readable storage medium may be provided in the form of a non-transitory storage medium. The term 'non-transitory storage medium' may mean a tangible device without including a signal, e.g., electromagnetic waves, and may not distinguish between storing data in the storage medium semi-permanently and temporarily. For example, the non-transitory storage medium may include a buffer that temporarily stores data.

[0385] In an embodiment of the disclosure, the aforementioned method according to the various embodiments of the disclosure may be provided in a computer program product. The computer program product may be a commercial product that may be traded between a seller and a buyer. The computer program product may be distributed in the form of a recording medium (e.g., a compact disc read only memory (CD-ROM)), through an application store (e.g., play store™), directly between two user devices (e.g., smart phones), or online (e.g., downloaded or uploaded). In the case of online distribution, at least part of the computer program product (e.g., a downloadable app) may be at least temporarily stored or arbitrarily created in a recording medium that may be readable to a device such as a server of the manufacturer, a server of the application store, or a relay server.

[0386] The embodiments of the disclosure have thus far been described with reference to accompanying drawings. It will be obvious to those of ordinary skill in the art that the disclosure may be practiced in other forms than the embodiments of the disclosure as described above without changing the technical idea or essential features of the disclosure. The above embodiments of the disclosure are only by way of example, and should not be construed in a limited sense.

Claims

1. A refrigerator comprising: a main body forming a storeroom; a thermoelectric element including a heater and a cooler, the cooler configured to cool air inside the storeroom; a fan configured to operate to bring air from outside the main body to the heater to be heated by the heater and exhaust the heated air to outside the main body; a filter configured to filter foreign materials from the air brought from outside the main body to the heater by operation of the fan; and a controller configured to notify that the filter is clogged in response to a rotation speed of the fan being higher than a reference rotation speed.
2. The refrigerator of claim 1, wherein the controller is further configured to: based on a lapse of a first period of time after the refrigerator is powered on, operate the fan for a second period of time, and based on the rotation speed of the fan during the second period of time, set the reference rotation speed.
3. The refrigerator of claim 2, wherein the controller is further configured to, based on an input indicating the filter is clean being received after notification that the filter is clogged, operate the fan for a third period of time.
4. The refrigerator of claim 3, wherein the controller is further configured to, based on a difference between the rotation speed of the fan during the third period of time and the reference rotation speed, determine whether the filter is clean.
5. The refrigerator of claim 4, wherein the controller is further configured to reset the reference rotation speed in response to determination that the filter is clean.
6. The refrigerator of claim 4, wherein the controller is further configured to notify that the filter is clogged in response to determination that the filter is not clean.
7. The refrigerator of claim 2, wherein the reference rotation speed includes: a first reference rotation speed, and a second reference rotation speed, and the controller is further configured to: operate the fan at a first level for the second period of time and based on the rotation speed of the fan operated at the first level, set the first reference rotation speed, operate the fan at a second level higher than the first level for the second period of time and based on the rotation speed of the fan operated at the second level, set the second reference rotation speed, notify that the filter is clogged in response to the rotation speed of the fan operated at the first level being higher than the first reference rotation speed, and notify that the filter is clogged in response to the rotation speed of the fan operated at the second level being higher than the second reference rotation speed.
8. The refrigerator of claim 7, wherein the controller is further configured to: based on an input indicating the filter is clean being received after notification that the filter is clogged, operate the fan at the first level for a third period of time and based on a difference between the rotation speed of the fan operated at the first level for the third period of time and the first reference rotation speed, determine whether the filter is clean, and operate the fan at the second level for the third period of time and based on a difference between the rotation speed of the fan operated at the second level for the third period of time and the second reference rotation speed, determine whether the filter is clean.
9. The refrigerator of claim 1, wherein the thermoelectric element is configured to operate so that a rate of temperature change of at least one of the heater and the cooler increases, and the controller is further configured to: notify that the filter is clogged in response to: the rotation speed of the fan being higher than the reference rotation speed, and after the thermoelectric element is operated, the rate of temperature change of the at least one of the heater and the cooler is smaller than a reference value.
10. The refrigerator of claim 1, further comprising: a communication interface configured to communicate with an external device, wherein the controller is further configured to transmit information regarding the filter being clogged to the external device through the communication

interface in response to the rotation speed of the fan being higher than the reference rotation speed.

11. A method of controlling a refrigerator including a main body forming a storeroom, a thermoelectric element including a heater and a cooler, the cooler configured to cool air inside the storeroom, a fan configured to operate to bring air from outside the main body to the heater to be heated by the heater and exhaust the heated air outside the main body, and a filter configured to filter foreign materials from the air brought from outside the main body to the heater by operation of the fan, and a controller, the method comprising: by the controller, notifying that the filter is clogged in response to a rotation speed of the fan being higher than a reference rotation speed.

12. The method of claim 11, further comprising: by the controller, based on a lapse of a first period of time after the refrigerator is powered on, operating the fan for a second period of time, and based on the rotation speed of the fan during the second period of time, setting the reference rotation speed.

13. The method of claim 12, further comprising: by the controller, based on an input indicating the filter is clean being received after notification that the filter is clogged, operating the fan for a third period of time.

14. The method of claim 13, further comprising: by the controller, based on a difference between the rotation speed of the fan during the third period of time and the reference rotation speed, determining whether the filter is clean.

15. The method of claim 14, further comprising: by the controller, resetting the reference rotation speed in response to determining that the filter is clean.
